

Isolated Reduction in Single-Breath Diffusing Capacity in Young, Healthy, Asymptomatic Women

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ABSTRACT: The clinical significance of an isolated reduction in the carbon monoxide diffusing capacity (DLCO) in nonsmoking, asymptomatic individuals is not known. Whether a reduced DLCO despite otherwise normal pulmonary function tests warrants further investigation remains unanswered. In this article, the authors describe five healthy, asymptomatic, young women who had isolated, reduced DLCO and subsequent follow-up examinations over a span of 6 years. This case series lends support to the contention that an isolated low DLCO in asymptomatic subjects is not clinically significant and does not necessitate additional medical inquiry. **KEY INDEXING TERMS:** Diffusing capacity; Asymptomatic women; Pulmonary function. [Am J Med Sci 1995;310(6):226-228.]

Carbon monoxide diffusing capacity (DLCO) was first described by Krogh in 1914.¹ It has since been used as a means of assessing gas transfer across the alveolar-capillary membrane. The primary disorders associated with a reduced DLCO include pulmonary emphysema, interstitial lung disease, and pulmonary vaso-occlusive disease.

In previous investigations, researchers focused on the value of an isolated reduced DLCO as an early diagnostic and/or prognostic indicator in clinical settings.^{2,3} What remains unclear is whether an isolated abnormal DLCO in a healthy, asymptomatic adult should alert physicians to early pulmonary pathology. In this article, we describe five subjects who had an isolated reduction in DLCO and underwent repeat studies during a span of 6 years.

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Materials and Methods

We followed five healthy, asymptomatic young women with low single breath diffusing lung capacity. These subjects were recruited to our laboratory as healthy control subjects. None of the subjects had concomitant respiratory, cardiac, or systemic disease, history of smoking, anemia, elevated carboxyhemoglobin levels, and/or obesity. All subjects had normal exercise tolerance and all participated in usual athletic activities.

Pulmonary function tests were repeated in four subjects. Spirometric studies were measured in triplicate, using a 12-L dry rolling seal spirometer. Predicted values of Schmidt et al⁴ were used for forced vital capacity and forced expiratory volume in one second. Lung volumes were measured using the helium dilution technique, and predicted values of Goldman and Becklake⁵ and Boren et al⁶ were used. The single-breath diffusing capacity was measured using a modified CPI (Cardiopulmonary Instruments, Texas), with manual valve in the sitting position in duplicate, by the single-breath method described by Ogilvie et al.⁷ The higher of the two acceptable maneuvers that agreed within 5% was reported. Linearity of the carbon monoxide analyzer was checked weekly. Breath-holding times during DLCO determinations were close to 10 seconds and were measured by the method of Jones and Mead.⁸ Carboxyhemoglobin levels were measured in these patients. Predicted values for DLCO were calculated using the data of Miller et al.⁹ Values outside the 95% confidence interval were considered abnormal. Absolute DLCO was corrected for hemoglobin using Cote's correction factor. However, because all patients had normal hemoglobin concentrations, the corrected DLCOs were virtually identical.¹⁰ Autopsy was performed in one subject who sustained an anaphylactic reaction to anesthesia during dental surgery, resulting in death.

Results

The characteristics of the study population are depicted in Table 1. The subjects (five, white women) were followed for 43 ± 32 months. Autopsy revealed normal lung tissue without any evidence of interstitial

Table 1. Pulmonary Physiologic Data

Subject	Age	Sex	F/U (months)	DLCO (mL/min/mmHg)	DLCO (%)	Hgb (gm/dL)	FVC (L) (% pred.)	FEV1 (L) (% pred.)	FEV1/FVC (%)	PaO ₂ (mmHg)
1*	29	F	77	17.8	59	14.6	3.99 (90)	3.76 (94)	94	-
1†				16.7	64	14.0	3.90 (91)	3.30 (93)	84	-
2*	38	F	46	14.9	61	14.2	4.28 (100)	3.57 (98)		84
2†					87	14.0	4.23 (96)	3.20 (96)	76	-
3*	25	F	10	20.2	64	14.0	4.06 (100)	3.40 (100)	83	92
3†				25.0	86	14.3	4.00 (100)	3.30 (100)	82	-
4*	27	F	3	16.8	61	14.0	4.53 (92)	3.74 (92)	82	97
4†			Died	-	-	-	-	-	-	-
5*	33	F	77	14.5	55	14.1	3.75 (92)	2.95 (88)	79	88
5†				21.3	84	13.9	4.05 (96)	3.04 (88)	75	-
Mean (±SD)*	30 (±5)		43 (±32)	16.8 (±2)	60 (±3)		4.12 (±.3)	3.62 (±.3)	84 (±5)	90 (±5)
Mean (±SD)†				20.9 (±3)	80 (±9)		4.05 (±1)	3.21 (±.1)	79 (±4)	

* Initial evaluation.

† Follow-up evaluation.

DLCO = diffusing capacity; FEV₁ = forced expired volume in one second; FVC = forced vital capacity; Hgb = hemoglobin; PaO₂ = arterial oxygen tension.

or vascular lung disease in subject 4. All subjects had normal spirometry and lung volumes, which were performed at the time of DLCO measurements. With the exception of subject 1, whose DLCO did not improve significantly, the other three subjects had a significant increase in DLCO. Three of these subjects had a normal DLCO after retesting. Alveolar volume in all patients remained virtually unchanged. Therefore, fractional change in inhaled versus exhaled carbon monoxide was solely responsible for changes in DLCO. All of the subjects were asymptomatic during initial testing and remained as such, with normal pulmonary examinations and exercise tolerance.

Discussion

This case series highlights five asymptomatic young subjects with low DLCO. Repeat DLCO levels were normalized in three, remained low in one, and in one subject who accidentally died, there was no evidence of parenchymal or vascular disease at autopsy. Use of DLCO in the evaluation and follow-up of patients with interstitial lung disease has been the subject of much debate.^{11,12} A reduction in the DLCO has been documented in many disease entities. Among obstructive lung diseases, emphysema and cystic fibrosis commonly are associated with a reduced DLCO. Gelb et al.¹³ found a reduced DLCO in clinically asymptomatic patients who had evidence of emphysema by lung biopsies, which were performed for other diagnostic reasons. Patients with acute or chronic pulmonary vaso-occlusive diseases often have low DLCO. Owens et al.¹⁴ found an isolated low DLCO with normal forced vital capacity in as many as 19% of patients with systemic sclerosis. In addition, the American Thoracic Society statement on the evaluation of impairment or disability secondary

to occupational lung disease has included DLCO as part of the initial evaluation to rate degree of impairment.¹² In several other studies, researchers addressed whether DLCO can predict exercise desaturation.^{15,16} Our group previously suggested that in evaluating exertional dyspnea, a normal DLCO does not require further invasive gas exchange testing, whereas an abnormal DLCO warrants further physiologic testing.¹⁵

The main determinants of DLCO are pulmonary capillary blood volume and the alveolar volume of the lung. Technical considerations play a role in accurate determination of this test. The accuracy of the single breath diffusing capacity in measuring the true overall pulmonary diffusion has been questioned.¹⁰ Although available evidence suggests that it is reasonably precise in healthy individuals, any inaccuracy in this measurement might increase the variability of the test beyond its true physiologic limits. In addition, single-breath DLCO has been dependent on alveolar oxygen tension, hemoglobin concentration, and the degree of lung inflation at the time of breathholding.^{10,17} These factors cannot be controlled completely in the clinical application of this test. However, in this series, all of our subjects had normal lung volumes and normal hemoglobin levels. There was no temporal relation between DLCO measurements and the subjects' menstrual cycle. The computational error or a break in the procedure, such as a leaky valve, was ruled out by careful step-by-step scrutiny.

Other physiologic factors influencing DLCO include age, race, body size and position, smoking, diurnal variation, and pregnancy. Carbon monoxide diffusing capacity is a function of body size, but DLCO appears to remain unchanged until weight to height ratio is greater than one.¹⁷ In our study, all subjects were white

and none of the subjects had a weight to height ratio greater than one. Carbon monoxide diffusing capacity tends to decrease with age, but this did not affect our study group, who were all young. Smoking results in decreasing DLCO independent of the effect on carboxyhemoglobin.¹⁷ Our study group did not include current or ex-smokers. Body position causes changes in DLCO, with an increase in DLCO from standing to sitting and from sitting to the supine position.¹⁰ All measurements in our study were obtained in the sitting position. Carbon monoxide diffusing capacity has been found to decline at a rate of 1–2% per hour from morning to night.¹⁸ Our measurements were done at various times, from 8 AM to noon.

In studies, researchers demonstrated that during pregnancy, DLCO increases in the first trimester, followed by a decline until 26 weeks of gestation, and without further changes until delivery.¹⁷ No pregnant females were included in this study. In another report, researchers found an isolated reduction in DLCO in patients with symptoms of influenza virus, which were not present in any of our patients.¹⁹ None of our subjects had any signs or symptoms of the flu or upper respiratory infection in the preceding 3 months of testing. Finally, with respect to prevalence of this problem, we are not aware of any systematic review of this issue. Therefore, we must ask whether a low DLCO in our subjects represents a normal variation and therefore a true physiologic state of the patients, because, by definition, a segment of the population will be "normal" and have a low DLCO.²⁰

In conclusion, we demonstrate in this case series that in healthy, young adults, an isolated reduced DLCO may not have clinical significance. It is possible that the low DLCO in our subjects is merely a normal variation. Alternatively, there may be other factors unknown to us that might have altered capillary blood volume that may explain the low DLCO and subsequent increase in three of our subjects.

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